# SEPARATEDNESS, UNRAMIFIEDNESS AND LOCAL QUASI-FINITENESS OF THE DIAGONAL READ OFF THE RELATIVE INERTIA

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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*Allowed standard prerequisites (editorial).*

- The construction of the relative inertia as a group algebraic space over a presentation of the source. Invoked inside the counted proof (source lines 1642-1647). Turns the identity section of the relative inertia into the identity of a group space.
- The statement that a group scheme over a base is separated exactly when its identity section is a closed immersion. Invoked inside the counted proof (source lines 1648-1651) as a cross-chapter pointer. Gives the equivalence of the three conditions in part (1).
- The description of the diagonal through the isomorphism spaces of a presentation. Used inside the counted proof (source lines 1655-1685). Identifies the pullbacks of the two diagonals with isomorphism spaces.
- The statement that the isomorphism space of a pseudo torsor is unramified or locally quasi-finite when the group space is. Invoked at the end of the counted proof (source lines 1686-1695) as a cross-chapter pointer. Transfers the properties from the inertia to the diagonal in parts (4) and (5).

*Source reference.* Original lemma lemma-inertia: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1220–1220. Navigation pointer only: the counted text refers to the Stacks item lemma-inertia, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Date*: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma lemma-isom-pseudo-torsor-aut-over-space: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1392–1392. Navigation pointer only: the counted text refers to the Stacks item lemma-isom-pseudo-torsor-aut-over-space, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original remark remark-inertia-is-group-in-spaces: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1292–1292. Navigation pointer only: the counted text refers to the Stacks item remark-inertia-is-group-in-spaces, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

*Source reference.* Original reference spaces-groupoids-lemma-group-scheme-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1658–1658. Navigation pointer only: the counted text refers to the Stacks item spaces-groupoids-lemma-group-scheme-separated, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

*Source reference.* Original reference spaces-groupoids-lemma-pseudo-torsor-implications: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1697–1697. Navigation pointer only: the counted text refers to the Stacks item spaces-groupoids-lemma-pseudo-torsor-implications, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

**Lemma 0.1.** *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks.*

(1) *The following are equivalent*
   (a) *$\mathcal{I}_{\mathcal{X}/\mathcal{Y}} \to \mathcal{X}$ is separated,*
   (b) *$\Delta_{f,1} = \Delta_f : \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ is separated, and*
   (c) *$\Delta_{f,2} = e : \mathcal{X} \to \mathcal{I}_{\mathcal{X}/\mathcal{Y}}$ is a closed immersion.*

(2) *The following are equivalent*
   (a) *$\mathcal{I}_{\mathcal{X}/\mathcal{Y}} \to \mathcal{X}$ is quasi-separated,*
   (b) *$\Delta_{f,1} = \Delta_f : \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ is quasi-separated, and*
   (c) *$\Delta_{f,2} = e : \mathcal{X} \to \mathcal{I}_{\mathcal{X}/\mathcal{Y}}$ is quasi-compact.*

(3) *The following are equivalent*
   (a) *$\mathcal{I}_{\mathcal{X}/\mathcal{Y}} \to \mathcal{X}$ is locally separated,*
   (b) *$\Delta_{f,1} = \Delta_f : \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ is locally separated, and*
   (c) *$\Delta_{f,2} = e : \mathcal{X} \to \mathcal{I}_{\mathcal{X}/\mathcal{Y}}$ is an immersion.*

(4) *The following are equivalent*
   (a) *$\mathcal{I}_{\mathcal{X}/\mathcal{Y}} \to \mathcal{X}$ is unramified,*
   (b) *$f$ is DM.*

(5) *The following are equivalent*
   (a) *$\mathcal{I}_{\mathcal{X}/\mathcal{Y}} \to \mathcal{X}$ is locally quasi-finite,*
   (b) *$f$ is quasi-DM.*

*Proof.* Proof of (1), (2), and (3). Choose an algebraic space $U$ and a surjective smooth morphism $U \to \mathcal{X}$. Then $G = U \times_{\mathcal{X}} \mathcal{I}_{\mathcal{X}/\mathcal{Y}}$ is an algebraic space over $U$ (Lemma lemma-inertia). In fact, $G$ is a group algebraic space over $U$ by the group law on relative inertia constructed in Remark remark-inertia-is-group-in-spaces. Moreover, $G \to \mathcal{I}_{\mathcal{X}/\mathcal{Y}}$ is surjective and smooth as a base change of $U \to \mathcal{X}$. Finally,

the base change of $e : \mathcal{X} \to \mathcal{I}_{\mathcal{X}/\mathcal{Y}}$ by $G \to \mathcal{I}_{\mathcal{X}/\mathcal{Y}}$ is the identity $U \to G$ of $G/U$. Thus the equivalence of (a) and (c) follows from Groupoids in Spaces, Lemma spaces-groupoids-lemma-group-scheme-separated. Since $\Delta_{f,2}$ is the diagonal of $\Delta_f$ we have (b) $\Leftrightarrow$ (c) by definition.

Proof of (4) and (5). Recall that (4)(b) means $\Delta_f$ is unramified and (5)(b) means that $\Delta_f$ is locally quasi-finite. Choose a scheme $Z$ and a morphism $a : Z \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$. Then $a = (x_1, x_2, \alpha)$ where $x_i : Z \to \mathcal{X}$ and $\alpha : f \circ x_1 \to f \circ x_2$ is a 2-morphism. Recall that

$$\begin{array}{ccc} Isom^{\alpha}_{\mathcal{X}/\mathcal{Y}}(x_1, x_2) & \longrightarrow & Z \\ \downarrow & & \downarrow \\ \mathcal{X} & \xrightarrow{\Delta_f} & \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \end{array} \quad \text{and} \quad \begin{array}{ccc} Isom_{\mathcal{X}/\mathcal{Y}}(x_2, x_2) & \longrightarrow & Z \\ \downarrow & & \downarrow{\scriptstyle x_2} \\ \mathcal{I}_{\mathcal{X}/\mathcal{Y}} & \longrightarrow & \mathcal{X} \end{array}$$

are cartesian squares. By Lemma lemma-isom-pseudo-torsor-aut-over-space the algebraic space $Isom^{\alpha}_{\mathcal{X}/\mathcal{Y}}(x_1, x_2)$ is a pseudo torsor for $Isom_{\mathcal{X}/\mathcal{Y}}(x_2, x_2)$ over $Z$. Thus the equivalences in (4) and (5) follow from Groupoids in Spaces, Lemma spaces-groupoids-lemma-pseudo-torsor-implications. $\square$